

Keeping DSE Capacity Aligned with Successive Model Generations

Internal working paper

The challenge

The Data Science Environment, DSE, is being asked to support a different workload from the one it was originally sized for. Successive model generations change the number and type of features, the historical data required, the volume processed and the way that processing has to be executed.

DSE operated with 50 workers until February 2026, when capacity increased to 100 workers. That was a significant uplift. The issue is that model-development demand continues to evolve after each capacity decision.

The core risk is a growing mismatch between model-generation workload and the compute available to execute it.

What changes from one model generation to the next

A new model generation is not simply a new version of the same workload. The processing profile can change materially even when the final model looks similar.

Workload difference	What changes	Impact on DSE
Feature volume	More candidate features are generated or evaluated	More calculations and larger intermediate outputs
Feature complexity	Heavier aggregations, joins, mappings or behavioural logic	Higher CPU and memory demand
Historical depth	Longer lookback periods are required	More source data must be read and processed
Data volume	Larger transaction or customer populations	Larger processing footprint
Execution pattern	Partitioning, reruns or multiple stages are needed	Longer elapsed time and more operational effort
Recurring demand	The established feature estate changes	The Weekly Model Plan carries a different ongoing workload

This is why feature count alone is not a sufficient measure of workload. FV9, for example, used approximately three years of historical data. Full-volume processing encountered crashes, so the workload had to be split into smaller parts and later consolidated.

Why the constraint keeps returning

Model-development workload changes continuously. DSE capacity changes at discrete points in time. If new model generations are introduced without reassessing the environment against the new processing profile, DSE is repeatedly asked to absorb work that was not part of the assumptions used when capacity was last set.

Capacity set	Workload evolves	DSE absorbs demand	Constraints appear	Next generation
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The evidence we need

The case should be built by comparing each model generation with the DSE capacity available at the time. The objective is to show whether capacity has kept pace with the workload, not simply whether capacity has increased.

Generation / period	DSE workers	Historical depth	Processing pattern	Observed constraint
OB4	TBD	TBD	TBD	TBD
FE8	TBD	TBD	TBD	TBD
FV9	50 / transition to confirm	~3 years	Partitioned	Full-volume crashes
Feb 2026	100	-	Capacity uplift	Worker capacity doubled
FV10	100	TBD	TBD	Current workload to establish

What the comparison should tell us

- Whether the current DSE has enough capacity and headroom for the workload now being introduced.
- Whether Feature Plans and recurring model processing need to be redesigned or optimised.
- Whether the sequencing or pace of future model generations should reflect the capacity available.

Required operating principle

DSE capacity planning should move with model-generation planning. Before a new generation is introduced, the expected processing profile should be understood against the capacity already in use.

This does not mean the answer is always more workers. The response may be additional capacity, a more efficient execution approach, a different model-generation cadence, or a combination of these.

Key message: DSE capacity has increased, but model-generation workload continues to change. If capacity, workload design and model-generation planning remain disconnected, the same constraints will continue to return.